

# The GridLab Beamer theme

A tour of what the theme gives you

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# What this deck is

- Every slide here shows a feature of `beamerthemegridlab.sty` and names the macro behind it.
- The title page you just saw, this title bar, the footer and the section dividers come from the theme — no setup in the `.tex` file.
- **Highlighted text** uses the accent colour; the theme ships GridLab blue and orange and both are one line to change.

The source of every slide is in `demo/demo.tex`.

**Slide furniture**

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## Blocks

Coloured title bar, light background.  
Group a definition, a result, an assumption.

- Bullets are accent-coloured
  - second level is an en-dash
    - ▶ third is a small triangle

## Footer

Short author, short title and frame / total sit at the bottom of every slide.  
Set them with `\renewcommand`  
`{\insertshortauthor}{F. Sossan}`.

## Corner logo

`\def\cornerlogo{logo.png}` puts a logo bottom-right of *every* slide.

A normal `itemize`:

- First point
- Second point
- Third point

`fitemize` spreads the items over the whole frame — handy when a slide holds three lines and looks empty:

- First point
- Second point
- Third point

`fenumerate` does the same for numbered lists.

\topicslide — a divider on demand

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- Every `\section{...}` automatically produces the full-colour divider you saw before this slide.
- `\topicslide{Title}` produces one wherever you want, without opening a section.
- `\titleslide{Title}` produces an extra title page mid-deck.

## Content macros

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Upright constants:  $\text{SOC}_{\max}$

A labelled underbrace, no wide gap:

$$\underbrace{P_t + Q_t}_{\text{apparent power}}$$

A forecast: *value at t, known at k*:  $\mathbf{P}_{k|t}$

Phasors:  $230/\underline{30^\circ}$       Cancelling:  $\cancel{x}$

```
 $\backslash\text{const}\{\text{SOC}\}_{\backslash\max}\$$   
 $\backslash[\ \backslash\text{uunderbrace}\{P_t + Q_t\}$   
          {apparent power}  $\backslash]$   
 $\backslash\text{vvector}\{P\}\{t\}\{k\}\$$   
 $230\backslash\text{phase}\{30^\circ\}\$$   
 $\backslash\text{cancel}\{x\}\$$ 
```

`mathtools`, `empheq`, `cancel` and `steinmetz` are loaded for you.

Quotations: “the grid is a queue with physics”

Open questions: **TODO: add the 2026 measurements**

One deck, several languages: This sentence is the English branch.

And this slide carries a watermark.

```
\say{the grid is a queue with physics}  
\todo{add the 2026 measurements}
```

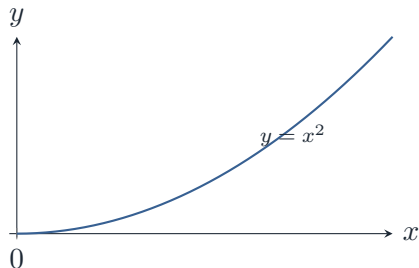
```
\def\EN{} % in the preamble  
\LANG{English}{Deutsch}{Francais}
```

```
\watermark{DRAFT}{8}
```

# Plots and diagrams

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## First-quadrant plot: q1simpleplot

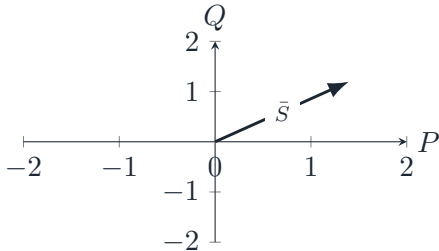


```
\begin{q1simpleplot}
  {0.95\textwidth}{0.5\textheight}
  {$x$}{$y$}
  \addplot[domain=0:1, samples=40,
    thick, color=primary]{x^2};
  \addplanelabel{0.62,0.5}
    {\scriptsize $y=x^2$}{anchor=west}
\end{q1simpleplot}
```

Axes through the origin, labels at the arrow tips, no ticks to read.

q1q2simpleplot does the same over  $x \in [-1, 1]$ .

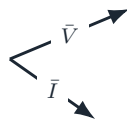
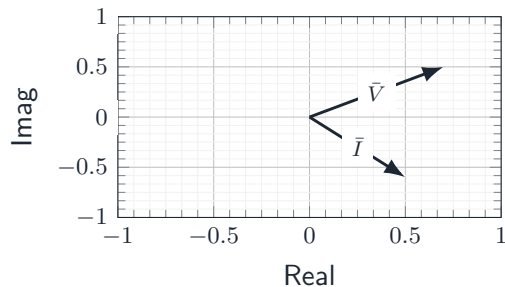
## General plot and vectors



```
\begin{plot}{w}{h}{xmin}{xmax}  
          {ymin}{ymax}{tickstep}  
          {$P$}{$Q$}  
  \vectorarrow{0,0}{1.4,1.2}{$\bar S$}  
\end{plot}
```

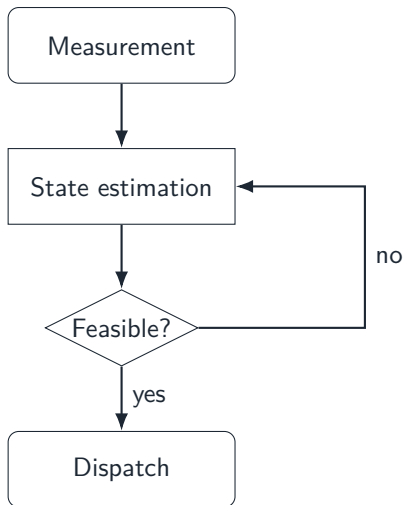
`\vectorarrow` draws an arrow between two axis coordinates and puts a white-backed label at its midpoint.

## Complex plane: phasorbox



Left: `phasorbox`, gridded with labelled axes. Right: `phasorbox_clean`, same but bare — for a diagram that should not look like a measurement.

## Flowchart styles



The styles startstop, process, decision and arrow come with the theme; TikZ and pgfplots are

**Making it yours**

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# Rebranding is two lines

```
\def\themeprimary{376092}    % main  
\def\themesecondary{ED7D31} % accent  
\usetheme{gridlab}
```

Set them *before* the theme is loaded and every title bar, divider, bullet and block follows.

```
\def\titlelogos{...}  
\def\titlebottomlogo{logo.png}  
\def\cornerlogo{logo.png}
```

Logos default to the ones shipped with the theme; override them, or set one empty to switch it off.

## Where to go next

- `../main.tex` — the template to start a real presentation from.
- `../README.md` — installing the theme, and the Overleaf route.
- `../gridlab/beamerthemegridlab.sty` — the theme itself; the macros above are all in the UTILITIES half of the file.

Thank you

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